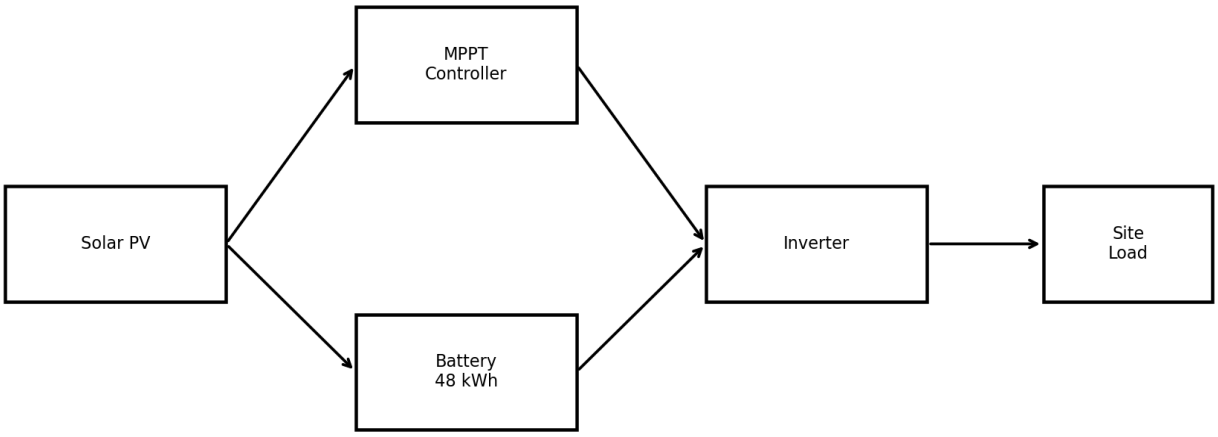
Harbor Ridge Renewable Microgrid

Technical Operations & Asset Reference Manual

Harbor Ridge Renewable Microgrid — Simplified Architecture



Document purpose: Technical reference for a fictional coastal microgrid containing solar PV, lithium-ion battery storage, power electronics, environmental sensors, and a small desalination load. This document intentionally mixes prose, tables, specifications, measurements, diagrams, charts, procedures, and maintenance data.

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1. Site Overview

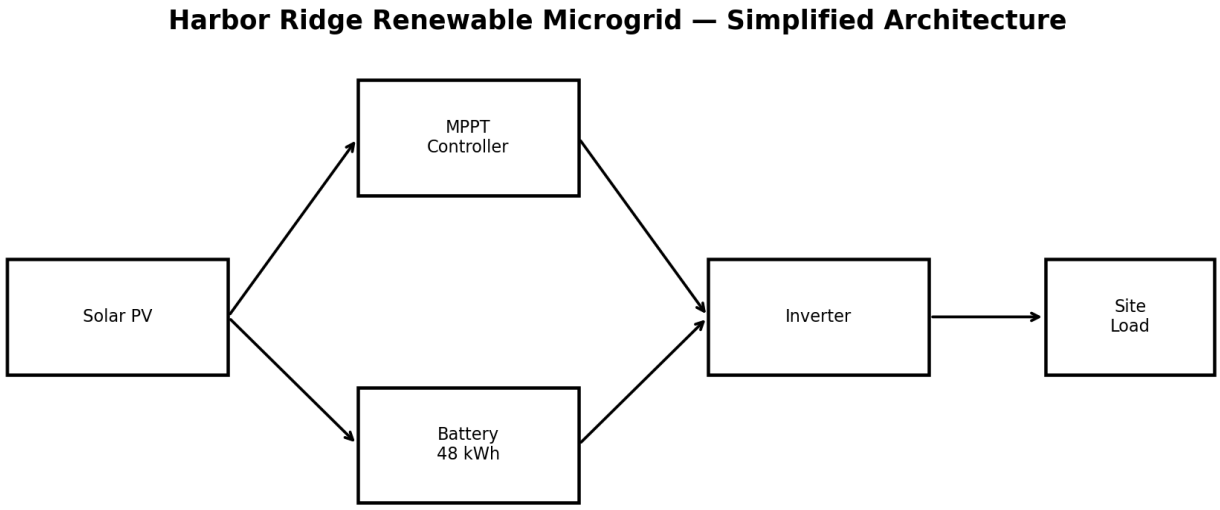
Harbor Ridge is a fictional 180 kW renewable microgrid serving a coastal research station. The electrical system combines rooftop and ground-mounted photovoltaic arrays with a 48 kWh lithium-iron-phosphate battery bank. The microgrid normally operates in grid-connected mode but can island during utility interruptions.

The site includes refrigeration, laboratory equipment, water pumps, lighting, communications equipment, and a variable-speed desalination pump. Salt spray, high humidity, and elevated ambient temperature are considered primary environmental stressors.

Site facts

Parameter	Value	Unit
PV nameplate capacity	180	kWdc
Battery usable capacity	48	kWh
Nominal battery voltage	51.2	V
Inverter rating	150	kWac
Average daily load	16.8	kWh/day
Peak site demand	112	kW
Elevation	36	m
Design ambient range	–5 to 46	°C
Design wind speed	42	m/s

2. Electrical Architecture



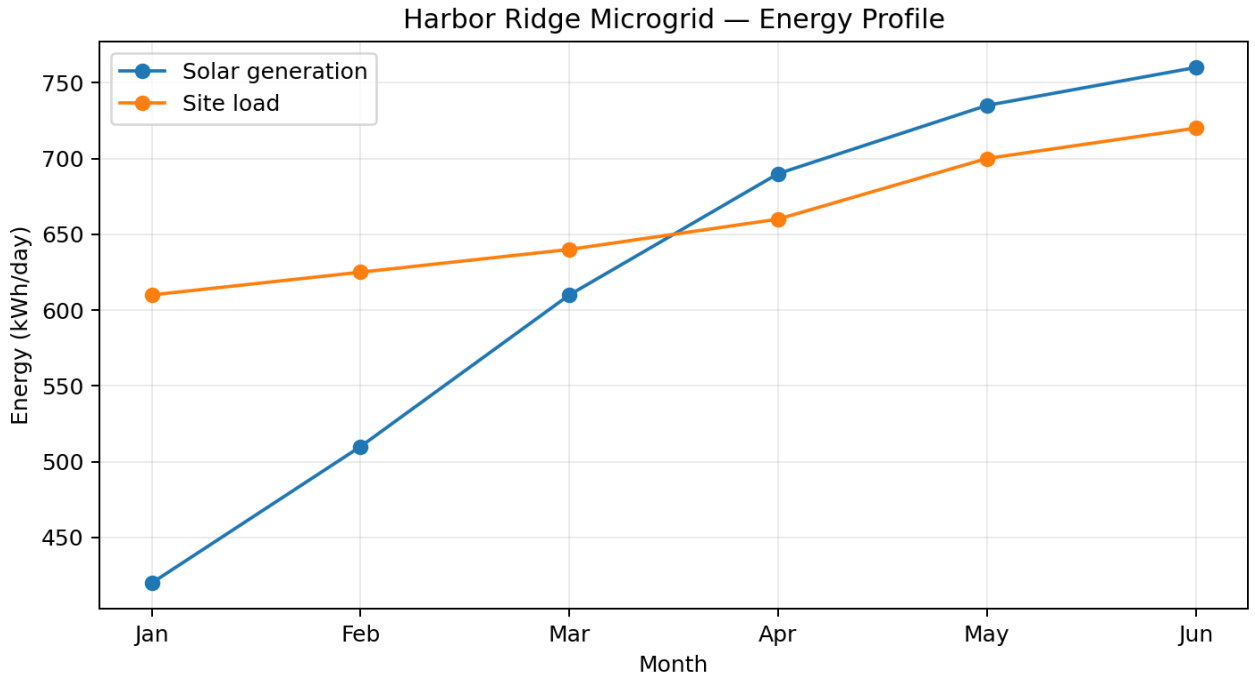
The photovoltaic strings feed an MPPT controller. DC power is distributed between the battery system and the inverter DC bus. The inverter supplies the AC distribution board, which feeds critical and non-critical site loads. A protection relay opens the utility intertie when islanding conditions are detected.

3. Major Equipment Register

Asset ID	Equipment	Model	Rating	Inspection interval
PV-01	PV Array A	HR-PV-60	60 kWdc	6 months
PV-02	PV Array B	HR-PV-60	60 kWdc	6 months

Asset ID	Equipment	Model	Rating	Inspection interval
PV-03	PV Array C	HR-PV-60	60 kWdc	6 months
BAT-01	Battery Rack	LFP-48-1000	48 kWh	3 months
INV-01	Grid-forming inverter	GX-150	150 kWac	6 months
PMP-04	Desalination pump	VS-45	45 kW	1 month
MET-01	Revenue meter	RM-400	Class 0.5	12 months

4. Six-Month Energy Profile



The chart compares average daily solar generation against site load. Generation increases during the modeled spring period while the site load remains comparatively stable.

5. Battery Operating Limits

Parameter	Normal	Warning	Trip / Protection
State of charge	20–90%	<20% or >90%	<10% or >95%
Cell temperature	10–35 °C	5–10 / 35–40 °C	<0 / >45 °C
Pack voltage	49.0–54.0 V	48.0–49.0 / 54.0–55.0 V	<48.0 / >55.0 V
Charge current	≤80 A	80–100 A	>100 A

Safety note: These values are fictional engineering test data and must not be used to configure a real battery management system.

6. Maintenance Procedure — PV Array Inspection

1. Isolate the relevant inverter and follow the site lockout/tagout procedure.
2. Inspect module surfaces for salt deposits, cracks, delamination, and discoloration.
3. Check cable glands and connectors for corrosion or mechanical damage.
4. Inspect mounting rails, fasteners, and grounding conductors.
5. Record string open-circuit voltage and compare with the previous inspection.

6 Clean modules only when approved water quality and weather conditions are available.

7 Upload photographs and measurements to the asset record.

7. Environmental Sensor Snapshot

Timestamp UTC	Temp °C	Humidity %	Wind m/s	Panel temp °C	Salt index
2026-10-08 06:00	21.4	81	4.2	23.1	0.18
2026-10-08 09:00	25.7	72	6.8	31.6	0.19
2026-10-08 12:00	30.2	64	8.1	46.2	0.23
2026-10-08 15:00	29.5	67	10.4	43.8	0.26
2026-10-08 18:00	25.1	74	7.2	29.4	0.27

8. Fault Code Reference

Code	Meaning	Likely cause	Recommended action
INV-E17	DC bus undervoltage	Low battery SOC or cable loss	Check SOC, fuses, and DC terminals
INV-E22	Grid frequency out of range	Utility disturbance	Verify utility frequency and relay state
BAT-W04	High cell temperature	Poor ventilation or high load	Reduce load and inspect cooling path
PV-W12	String mismatch	Shading, connector issue	Compare string voltage/current
PMP-E03	Pump overload	Blocked intake or bearing issue	Stop pump and inspect mechanically

9. Technical Notes

The modeled battery round-trip efficiency is 91.5%. A simplified energy balance for a daily cycle is $E_{out} = \eta \times E_{in}$, where $\eta = 0.915$. For a 32.0 kWh charge event, the expected delivered energy is approximately 29.28 kWh before auxiliary consumption.

The control system uses thresholds with hysteresis to avoid rapid state transitions. For example, the desalination pump may start when battery SOC exceeds 55% and stop when SOC falls below 45%, provided grid and water-tank conditions are satisfied.

10. Reference Questions

1 What is the total PV nameplate capacity?

2 Which equipment has a 48 kWh rating?

3 What is the battery warning temperature range?

4 Which fault code relates to a blocked pump intake?

5 At what time was the highest panel temperature recorded?

6 What was the wind speed at 15:00 UTC?

7 Compare solar generation and site load in June.

8 What inspection interval applies to the revenue meter?

9 Explain the purpose of the MPPT controller using the architecture diagram.

10 Calculate the expected delivered energy from a 32.0 kWh charge event using $\eta = 0.915$.

END — FICTIONAL TEST DOCUMENT / NOT FOR REAL-WORLD ENGINEERING USE